УНИВЕРСИТЕТ ИТМО

Технологии машинного обучения Machine learning technologies

Lecture 3 – The basics of supervised learning

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ITMO University

Supervised Algorithms overview

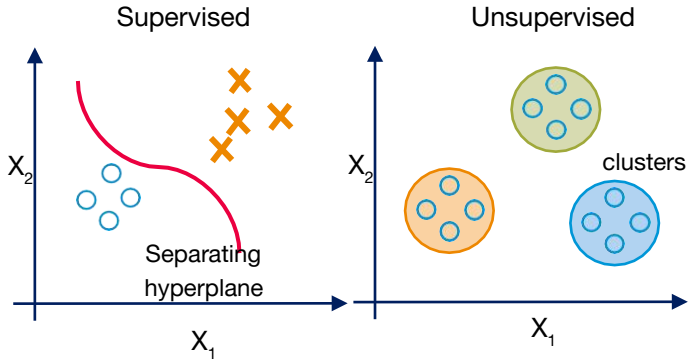
Supervised and unsupervised learning

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Supervised Learning - each data sample is an object-label pair. It is required to find the functional dependence between the label and object's features and to construct an algorithm that takes the description of the object as input (features, characteristics) and produces the response (prediction) as the output.

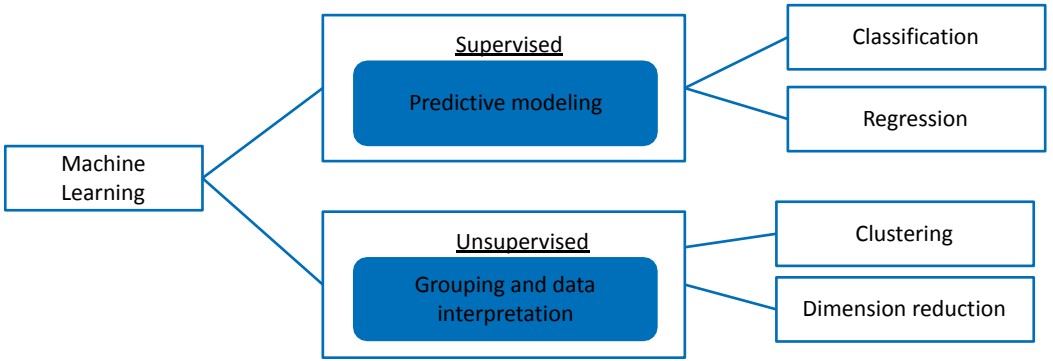
Regression makes a forecast based on the characteristics of the data sample. The output should be **a real number**.

Classification - produces **categorical response** based on the set of the features.



Unsupervised learning - no labels(answers) are given, and you need to look for dependencies between objects "internally".

Clustering is an example of unsupervised learning. The goal is distribute data into groups.



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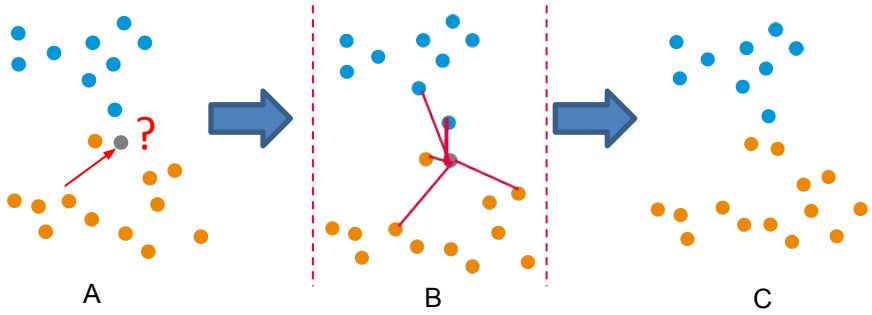
K-nearest neighbours

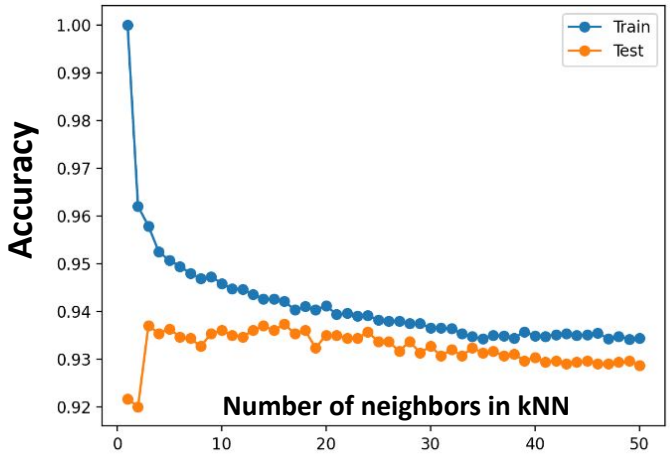
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kNN algorithm:

To classify each of the objects of the test sample, you must sequentially perform the following operations:

1. calculate the distance to each of the objects of the training sample;
2. select **k** objects of the training sample, the distance to which is minimal;
3. the class of the classified object is the class most frequently encountered among the **k nearest neighbors**.



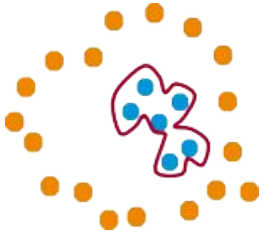


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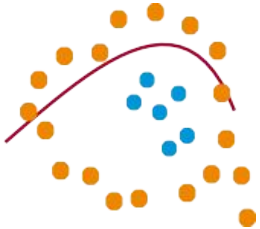
Overfitting

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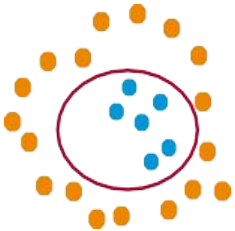
Overfitting



Underfitting

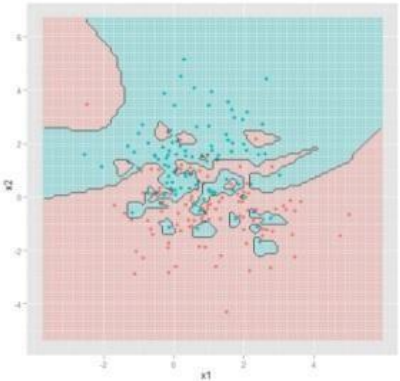


Optimum

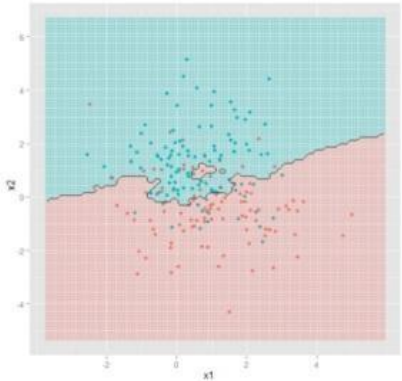


Overfitting is a phenomenon associated with too strong fitting of the model for the training set, leading to a highly biased prediction estimate.

Regularization is a way to reduce the complexity of a model to prevent overfitting.



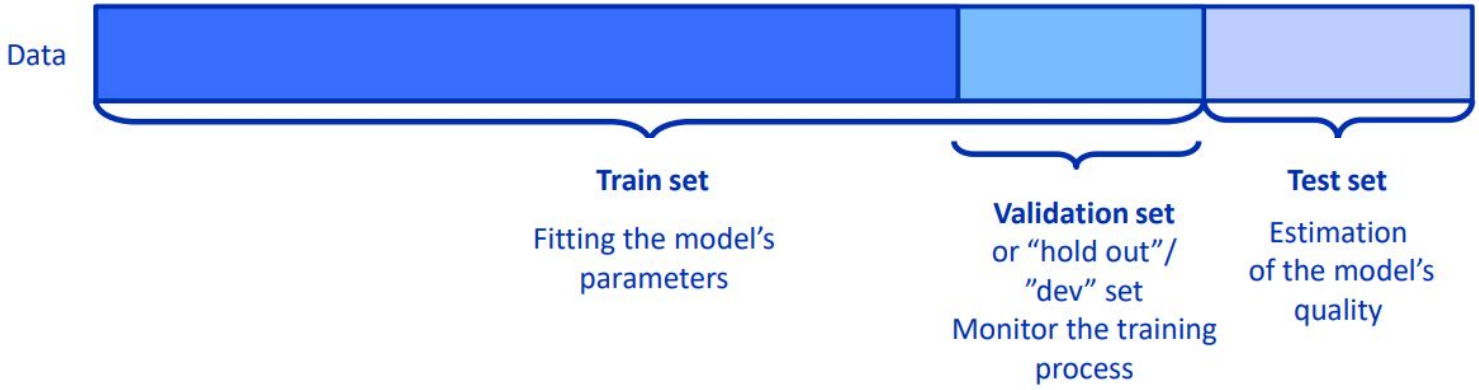
K=1



K=15

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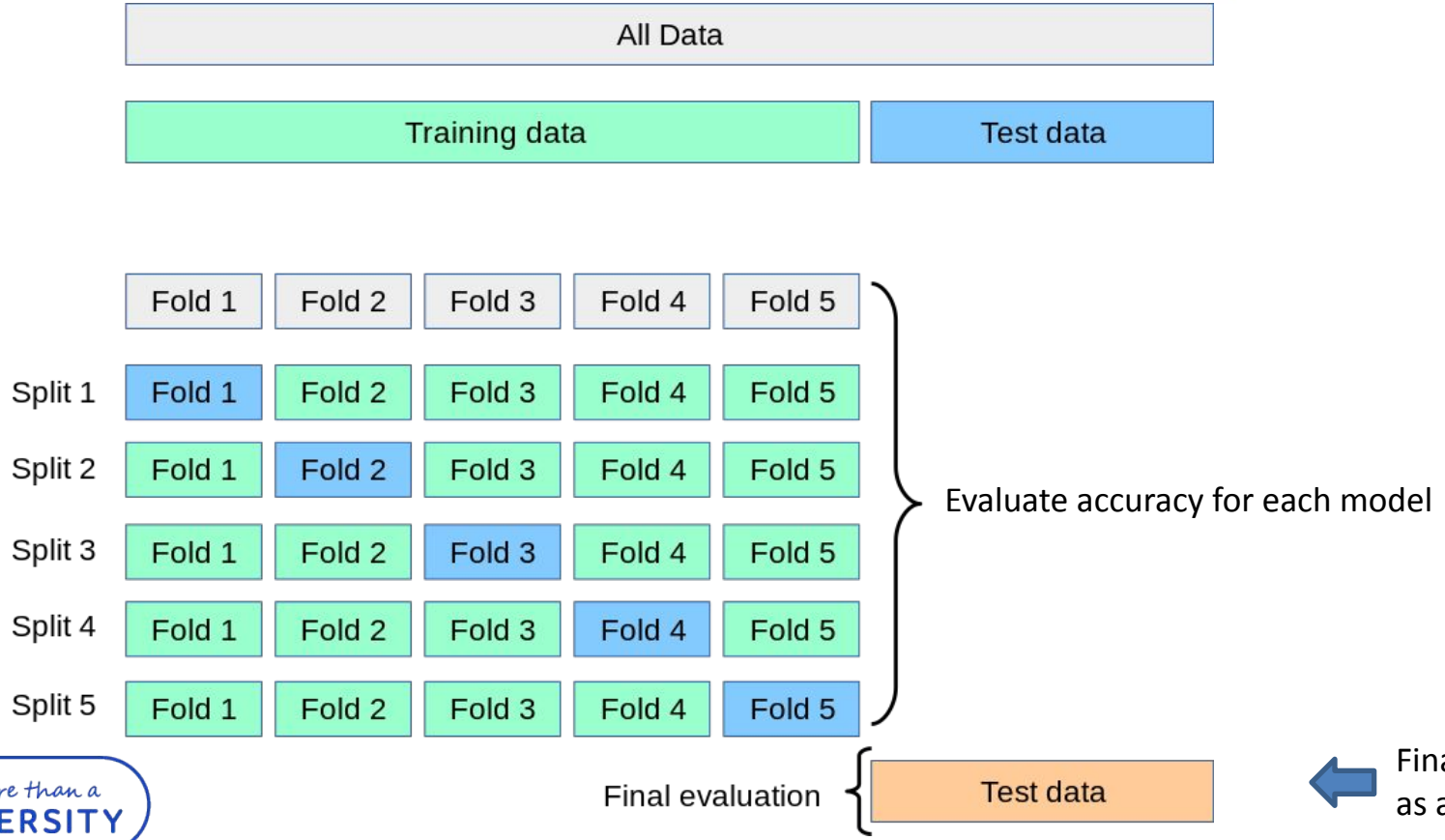
Train/test split



	Dataset size	Train set	Dev set (hold out)	Test set
"small"	100 – 10 000	70%±10%	15%±10%	15%±10%
"large"	> 1 000 000	98%	0.5%-1%	0.5%-1%

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2,
random_state=42, stratify=y )
```

Cross-validation

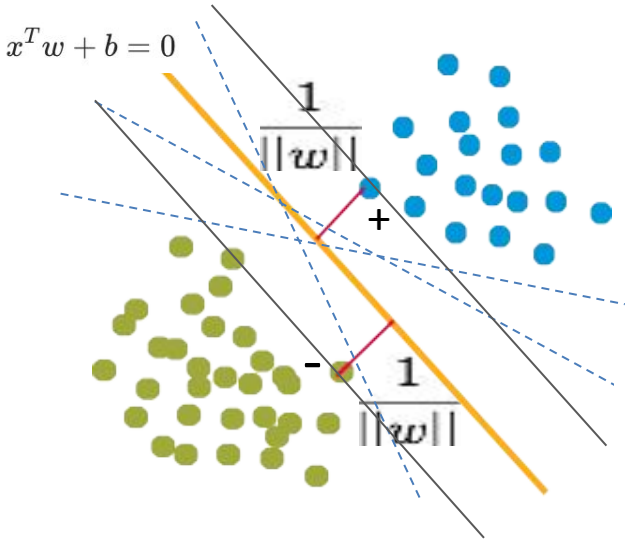


SVM algorithm

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Support Vector Machine

- The main idea is to look for a separating hyperplane with a maximum gap (margins) in the multidimensional space of data points.
- If the sample is linearly separable, then there exists such a hyperplane, the indent from which to each object is positive (orange line).
- Such samples are almost never exist in practice; in this case, it is necessary to relax the restrictions (1-ξ), allowing some objects to be placed on the separable width or on the "territory" of another class.



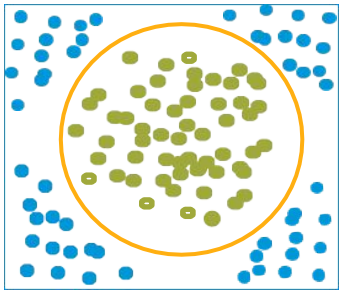
It can be shown that the width of separating area:

$$\left\langle (x_+ - x_-), \frac{w}{\|w\|} \right\rangle = \frac{2}{\|w\|}$$

Prediction: $\hat{y}_i = \text{sign}(x_i^T w + b)$

Optimization problem:

$$\begin{cases} \frac{1}{2}\langle w, w\rangle + C\sum_{i=1}^{\ell}\xi_i \to \min\limits_{w, w_0, \xi}, \\ y_i(\langle w, x\rangle - w_0) \geq 1 - \xi_i, \qquad i = 1, ..., \ell, \\ \xi_i \geq 0, \qquad i = 1, ..., \ell. \end{cases}$$



If the sample is not linearly separable?

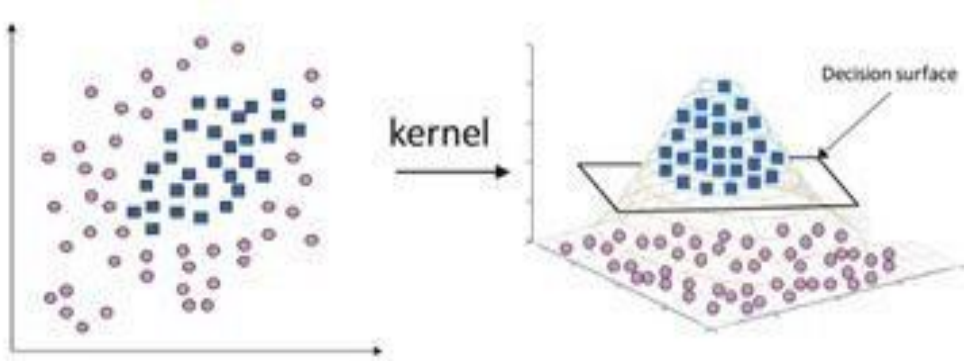
- all elements of the training sample are embedded in the space ***X*** of a higher dimension using a special mapping ***φ().***
- Mapping ***φ()*** is chosen so that *in the new space **X** the sample is linearly separable.*

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Kernel trick for SVM

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- **Kernel Trick** allows you to use linear equations but with construction of separating hyperplane **in a higher dimensional space**.
- The kernel trick offers an efficient way to transform data into higher dimensions.



Initial data (2D)

Data after transform

In practice, the following kernels are found:

- **Polynomial;**
- **Radial basis function;**
- **Gaussian radial basis function;**
- **Sigmoid.**
- **other**

Optimization problem:

$$\begin{cases} \frac{1}{2}\langle w, w\rangle + C\sum_{i=1}^{\ell}\xi_i \to \min\limits_{w, w_0, \xi}, \\ y_i(\langle w, x\rangle - w_0) \geq 1 - \xi_i, \qquad i = 1, ..., \ell, \\ \xi_i \geq 0, \qquad i = 1, ..., \ell. \end{cases}$$

$$x \to \varphi(x), \quad w \to \varphi(w), \quad \langle w, x\rangle \to \langle \varphi(w), \varphi(x)\rangle.$$

$$K(w, x) = \langle \varphi(w), \varphi(x)\rangle$$

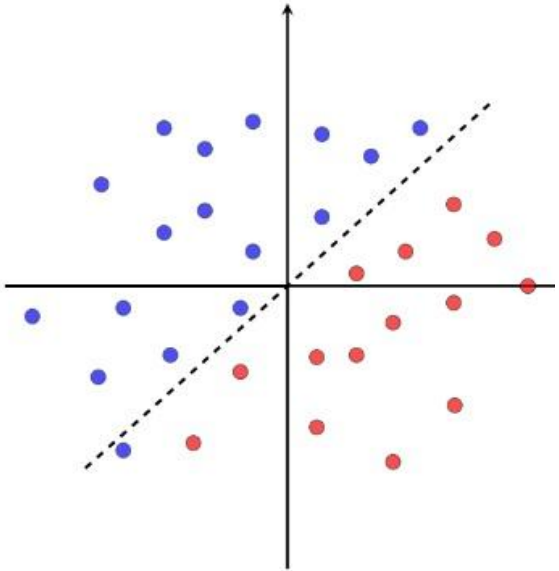
K(x,y) — kernel.

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Kernel trick for SVM

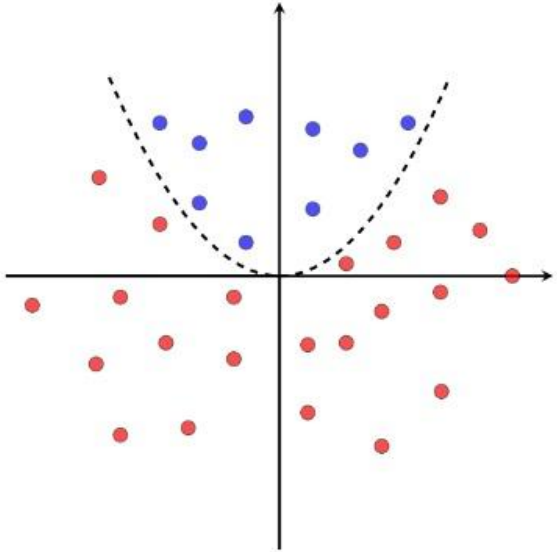
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$$K(w,x) = \langle w,x \rangle.$$



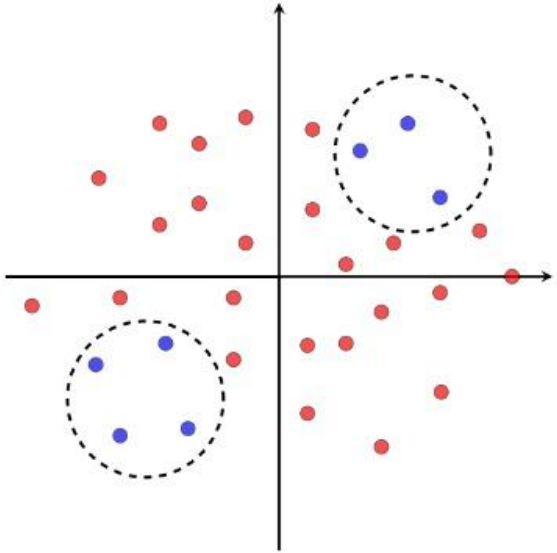
Linear kernel

$$K(w,x) = (\langle w,x \rangle + r)^d.$$



Polynomial kernel

$$K(w,x) = \exp\left(-\gamma \|w-x\|^2\right).$$



Radial basis function (RBF) kernel

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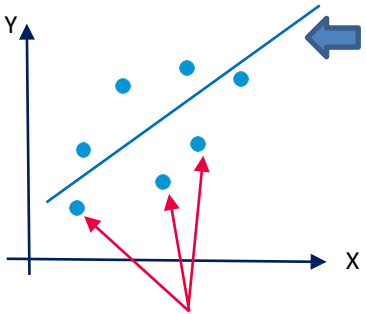
Linear regression

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Linear regression is a method of restoring the linear dependence of one (dependent) variable on another or several other variables (regressors).

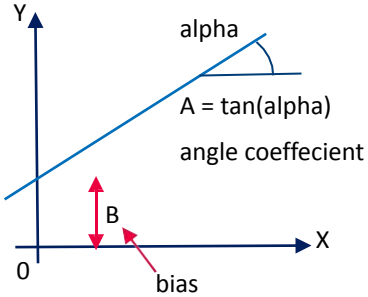
The label of the object is continuous.

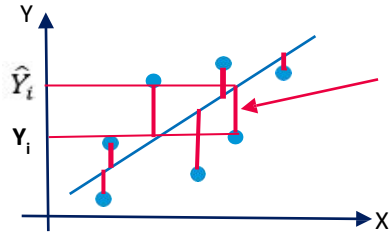
Model: linear regression $y = w^T x + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$



Train sample (***X***), «right answer» (***Y***)

Regression model ***Y*** = ***Ax+B***





Error is the difference between real and predicted values

Metrics of accuracy

- **(Mean Absolute Error)** $MAE = \frac{1}{n}\sum_i |Y_i - \hat{Y}_i| \to min$
- **(Mean Squared Error)** $MSE = \frac{1}{n}\sum_i (Y_i - \hat{Y}_i)^2 \to min$
- **(Root Mean Squared Error)** $RMSE = \sqrt{MSE}$ Transform to the same Units of measurement, as for Y
- **R-squared (determination coefficient)** $R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \to 1$, $SS_{res} = \sum_i (Y_i - \hat{Y}_i)^2$ Unexplained variance of Y; $SS_{tot} = \sum_i (Y_i - \bar{Y})^2$ Variance of Y

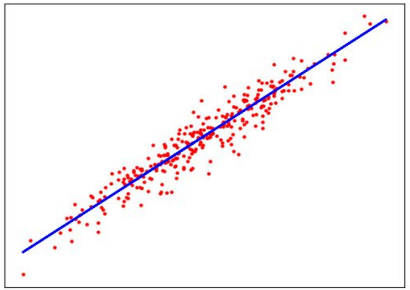
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Linear and Logistic Regression

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Linear Regression - the target is continuous. We have a matrix, where rows correspond to specific objects with features and target y – responses.

X_1	X_2	X_3	Y
11	68.028164	0	21.46126
12	69.086446	0	23.28792
13	84.806730	1	18.71906
14	19.011313	1	18.50209
15	63.046323	1	22.37717
16	82.686964	1	24.19955
17	59.263664	1	20.64198
18	88.756598	0	29.50144
19	77.884304	1	24.49684
20	9.346073	0	27.15191



Model: linear regression $y = w^T x + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$

The least squares method allows you to find a vector *w* for a known matrix X and a known target vector y:

$$\arg\min_w Q(w) = \frac{1}{2N} \sum_{i=1}^{N} \left(x_i^T w - y_i\right)^2 = \frac{1}{2N} \|Xw - y\|_2^2$$

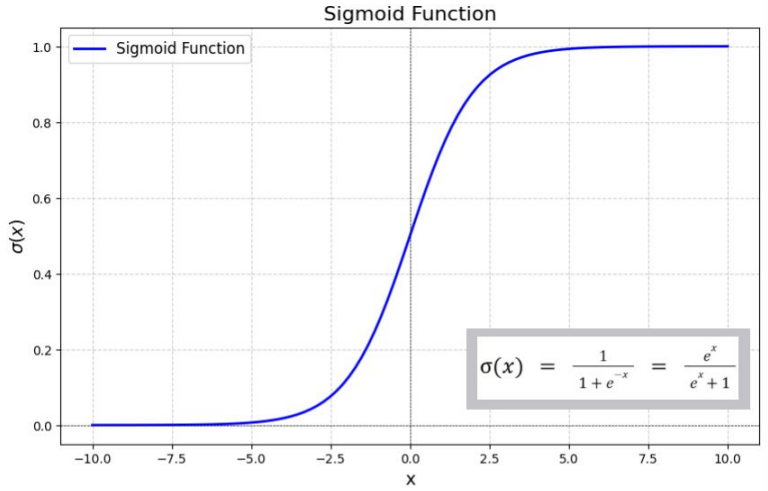
Logistic regression is a *categorical* type response model.

Linear regression evaluates a value in a form of $Y = w^T x + b$, which is linear equation.

Logistic regression **applies non-linearity** in a following form:

$$Y = \frac{1}{1 + e^{-(w^T x + b)}}$$

- Sigmoid ("step") activation function



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Weights can be found f.e., via gradient descent algorithm.

Logistic regression

Logistic loss function ("log-loss"/”binary cross-entropy”) - determines the average difference between the probabilities of assigning objects to classes 1 and 0 and their actual labels:

$$log_loss(actual, predicted) = -\frac{1}{n}\sum_{i=1}^{n}(actual_i \cdot \log(predicted_i) + (1 - actual_i) \cdot \log(1 - predicted_i))$$

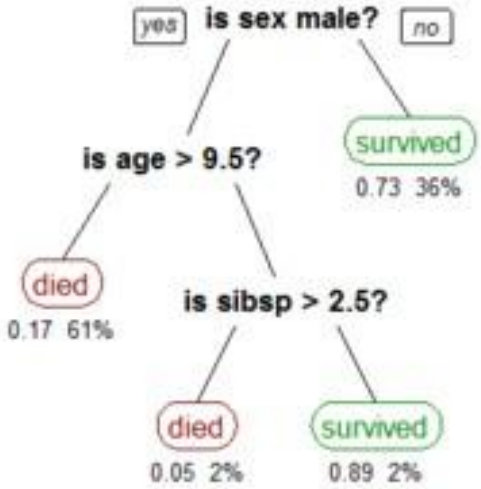
Logical Classifiers

Logical / metric classifiers

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Goal for **metric classifiers**: to find **weights of objects** within decision rule or the distance between them (object-based)

Goal for **logical classifiers**: to find **feature-based predicates** as parts of decision rule (feature-based)



1. Medicine

if (AGE > 60) and (WAS_HEART_ATTACK == TRUE)

OPERATION = FALSE (high risk)

2. Scoring models

if (SALARY == “low”) and (OVERDUE_CREDITS.COUNT > 0)

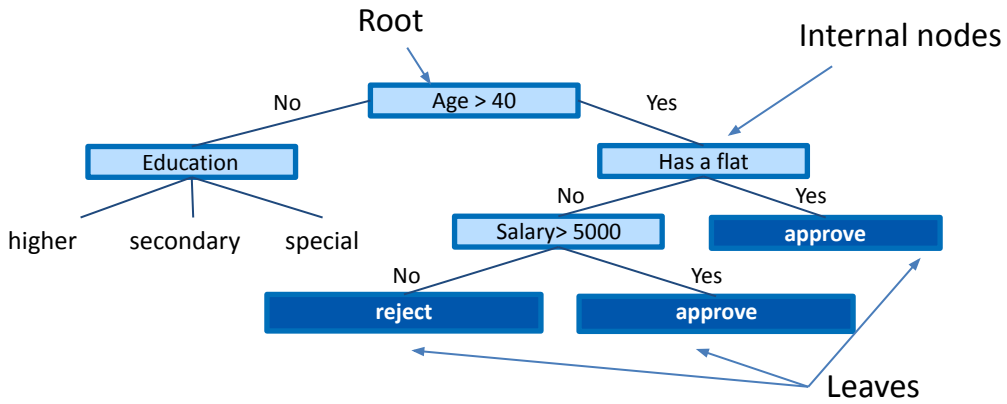
DECISION = false (high risk)

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Decision trees

Decision tree:

- the algorithm for constructing a decision tree consists in sequentially splitting the sample in a greedy way into two (or more) parts until each part contains objects of only one class.
- division is performed by a predicate (condition) that is selected from a set of elementary predicates.
- in practice, simple threshold conditions for the value of a feature are most often taken as elementary predicates.



To choose the condition – we need to choose most optimal way of **impurity decreasing** in the in current node:

Impurity for data sample X for the feature J

$$Q(X_m, j, t) = \frac{|X_\ell|}{|X_m|} H(X_\ell) + \frac{|X_r|}{|X_m|} H(X_r)$$

Criterium of information gain

Examples of measures of distribution's impurity are:

Entropy: $\Phi(U) = -\sum_{i}^{N} p_i \log_2 p_i$

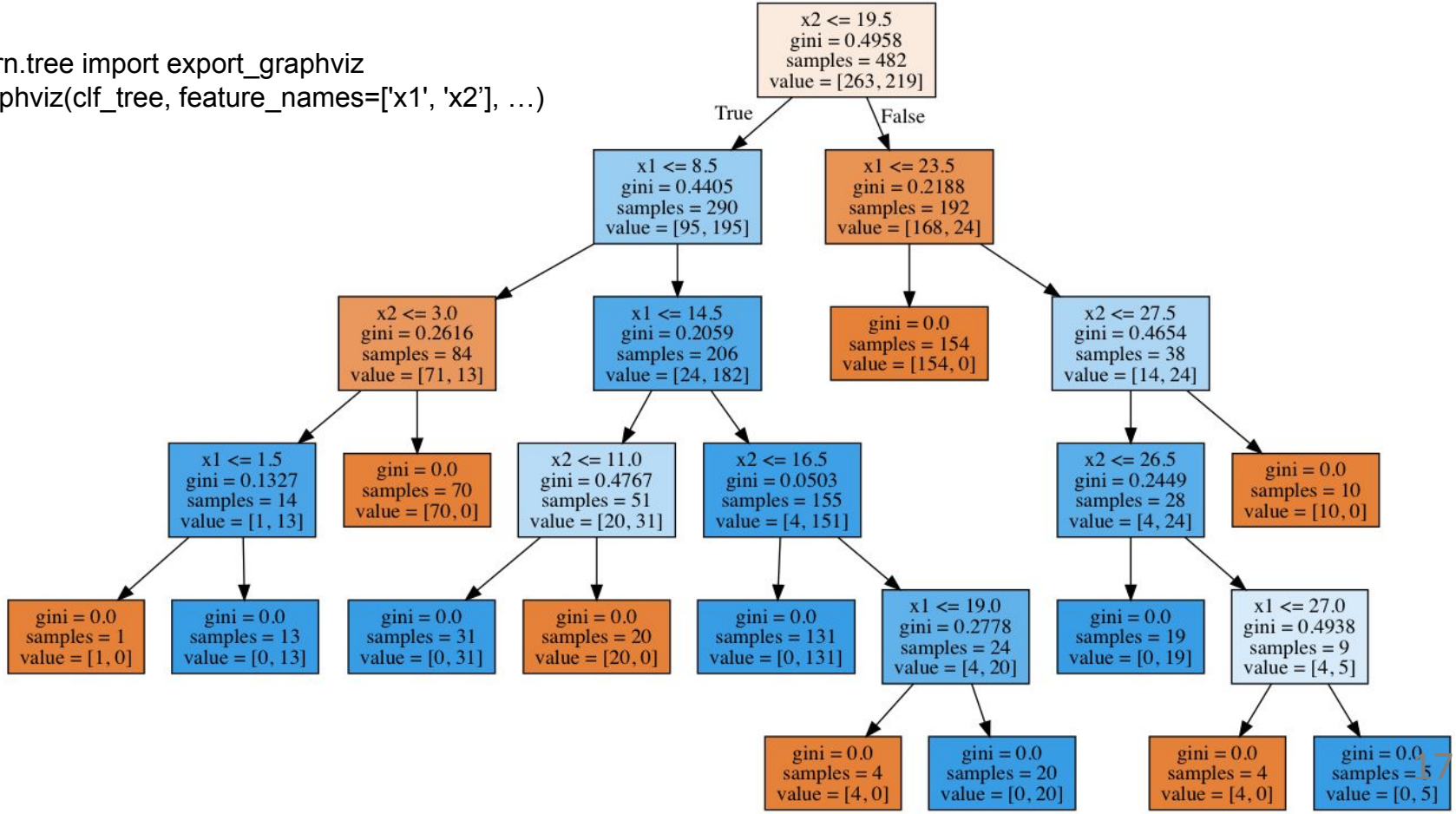
- where N is the number of possible classes, and p_i - is the probability of an object belonging to the i-th class.

Gini criterion: $\Phi(U) = -\sum_{i \neq j} p_i p_j = \sum_{i=1}^{N} p_i (1 - p_i)$

Decision trees

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```
from sklearn.tree import export_graphviz
export_graphviz(clf_tree, feature_names=['x1', 'x2'], …)
```



Ensembles

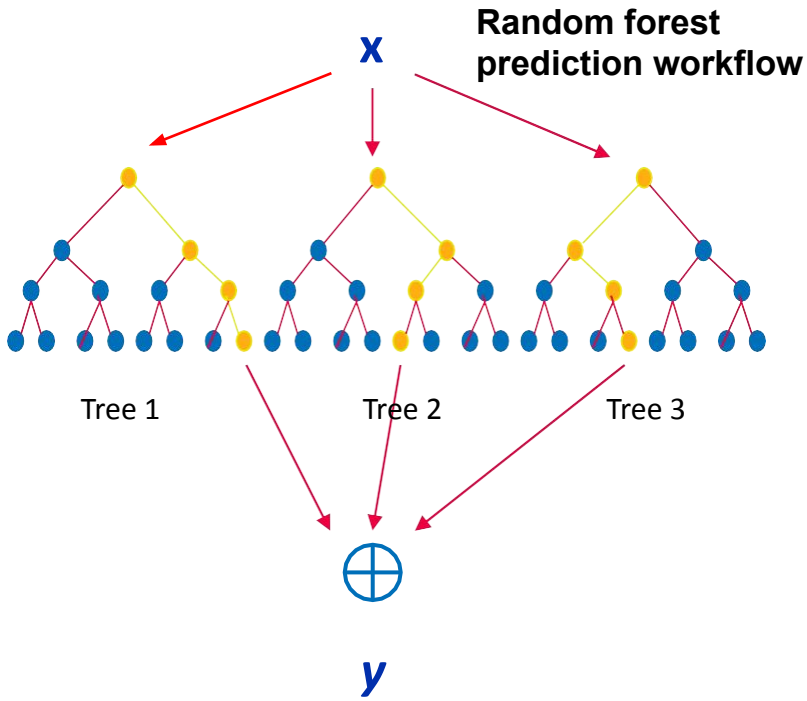
Random forests

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The **Random forest** is one example of the ensemble of classifiers.
"Bagging over decision trees"

Algorithm:

- M samples of size N are generated for each classifier from initial dataset - “Bootstrap sampling”
- Each elementary classifier (each algorithm defined on its subspace) is independently trained. For each split there are chosen specific set of features.
- The classification of the main sample is performed on each of the subspaces (also independently).
- the final decision is made on the belonging of the object to one of the classes:
- for the **classification problem,** we choose a solution for most of the results given by the classifiers – **“Majority voting”**;
- in the **regression problem** - by their **average value**.



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General Gradient Boosting Algorithm

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$$a(x_i) + b(x_i) = y_i, \quad i \in \{1, 2, \ldots, m\}$$

a(x) - are the predictions of some model. Consider correction condition for a(x): a(x)+b(x)≈y

$$F = \sum_{i=1}^{m} L(y_i,\ a(x_i) + b_i) \ \rightarrow \ \min_{(b_1, \ldots, b_m)}$$

Objective function: minimize loss

$$-(L'(y_1, a(x_1)),\ \ldots,\ L'(y_m, a(x_m)))$$

Antigradient direction of F

$$b_i = -L'(y_i,\ a(x_i)), \quad i \in \{1, \ldots, m\}$$

So it is optimal to set b_i to values of antigradients

$$(x_i,\ -L'(y_i,\ a(x_i)))_{i=1}^{m}$$

New training set on which b(x) is fit: antigradients are new labels

$$a_{t+1}(x) = a_t(x) + b_t(x)$$

Ensemble update rule (primitive algorithm)

$$a_{t+1}(x) = a_t(x) + \eta_t \cdot b_t(x) = \eta_1 \cdot b_1(x) + \ldots + \eta_t \cdot b_t(x)$$

Usually the “learning rates” are added to have linear combination of b(x)

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Source: A. Dyakonov, «Градиентный бустинг» (pp. 1–2), alexanderdyakonov.wordpress.com — — marks derivation steps left unnumbered in the source.

Popular Boosting Algorithms

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XGBoost (Extreme Gradient Boosting)

- Gradient boosting with decision trees.
- Sequentially adds trees to minimize a differentiable loss function using gradient information.
- High predictive accuracy, regularization to reduce overfitting, efficient training, parallelized tree construction.
- Best for Structured/tabular data and complex nonlinear relationships.
- Requires hyperparameter tuning; categorical features typically need encoding.

AdaBoost (Adaptive Boosting)

- Boosting that combines weak learners, often shallow decision trees (decision stumps).
- **Increases the weights** of misclassified training examples so subsequent learners focus on harder cases.
- Simple, relatively interpretable, can improve weak learners substantially.
- Best for: Relatively clean datasets with simple decision boundaries.
- Limitations: Sensitive to noise and outliers because it emphasizes misclassified examples.

CatBoost (Categorical Boosting, originally developed by Yandex)

- Gradient boosting with decision trees.
- Uses ordered boosting and specialized techniques for categorical features to reduce target leakage and prediction bias.
- Handles categorical features natively, often works well with minimal preprocessing, helps reduce overfitting.
- Best for: Tabular datasets with many categorical variables.
- Limitations: Can be computationally expensive; training may be slower than some alternatives in certain settings

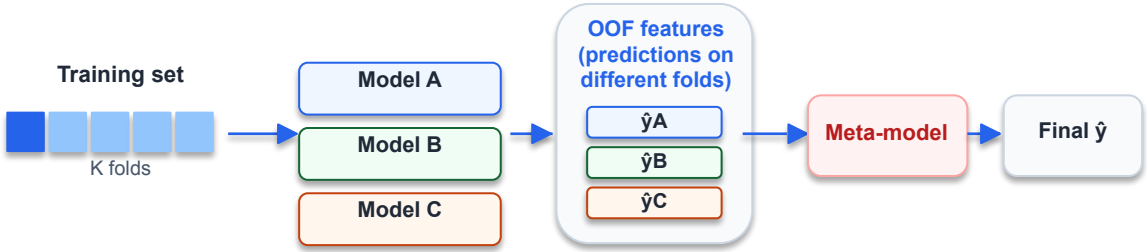
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Stacking and Blending

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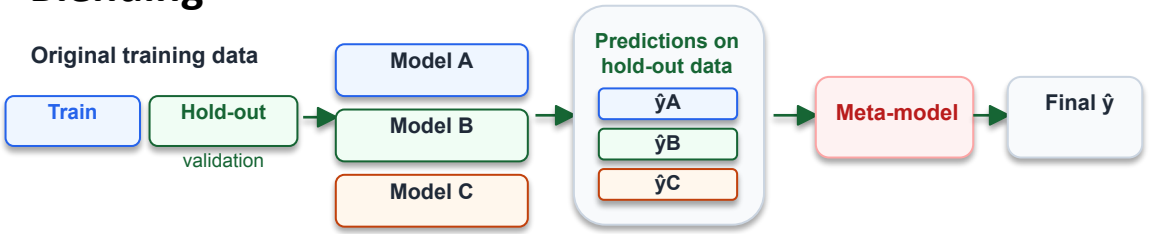
Two ensemble techniques combine base-models predictions to improve predictive performance using meta-model.

Stacking



Split training data into K folds (cross-validation)
Train on K−1 folds; predict the held-out fold by each of the model
Collect out-of-fold predictions as new dataset;
Train meta-model.

Blending

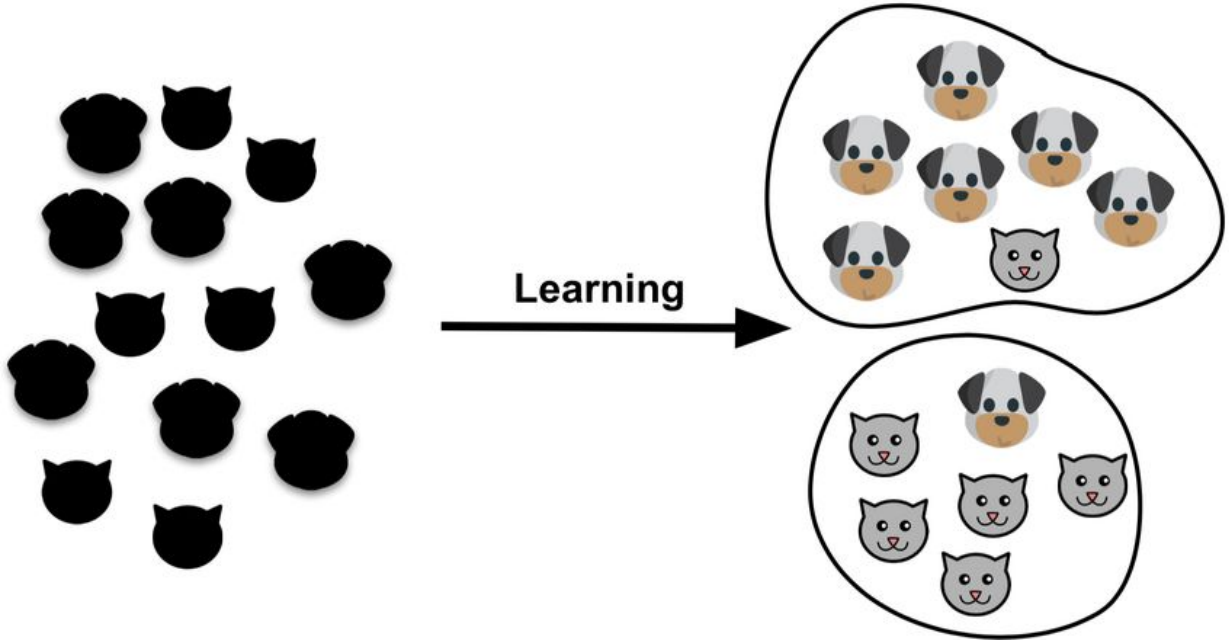


Split data into base-training and validation sets.
Train base models on train data;
Predict on validation data.
Train meta-model on validation predictions.

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Classification metrics

Binary classification accuracy metrics



- How good is our model?
- Is it good in classifying cats or dogs? Or both?

Binary classification accuracy metrics

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«Confusion matrix»

Actual \ Predicted	0	1
0	TN (True Negative)	FP (False Positive)
1	FN (False Negative)	TP (True Positive)

An error of the **I type** is an error consisting in making a false hypothesis.

(False Positive) - “False Triggering".

An error of the **II type** is an error consisting in the rejecting of a correct hypothesis

(False Negative) – “Missing a goal."

Accuracy - shows the percentage of correct classifications.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Recall - shows the proportion of the correctly found 1 class objects to the total number of real 1 class objects.

$$Recall = \frac{TP}{TP + FN}$$

Precision - shows the proportion of correctly found 1 class objects among the objects selected by the classifier as 1 class.

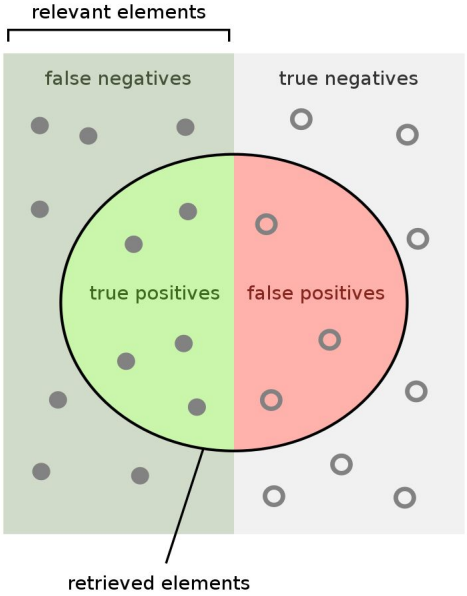
$$Precision = \frac{TP}{TP + FP}$$

For a general assessment of the quality of the classifier, the **F1-measure** is often used - the harmonic mean between precision and recall:

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

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Binary classification accuracy metrics



How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{retrieved elements}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{relevant elements}}$$

Binary classification accuracy metrics

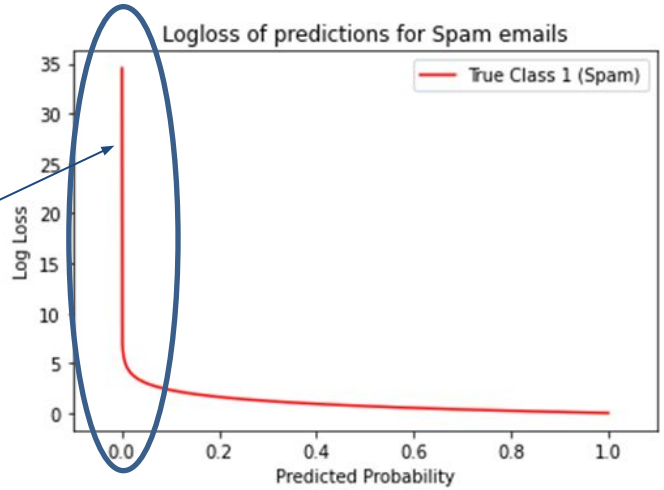
УНИВЕРСИТЕТ ИТМО

Logistic loss function ("log-loss"/"binary cross-entropy") - determines the average difference between the probabilities of assigning objects to classes 1 and 0 and their actual labels:

$$log_loss(actual, predicted) = -\frac{1}{n}\sum_{i=1}^{n}(actual_i \cdot \log(predicted_i) + (1 - actual_i) \cdot \log(1 - predicted_i))$$

It is sensitive to the wrong predictions with high (in absolute values) errors.

if there is wrong FN predictions



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ROC-AUC

УНИВЕРСИТЕТ ИТМО

ROC-AUC –
“Receiver Operating Characteristics (ROC) – Area-Under-the-Curve (AUC)”

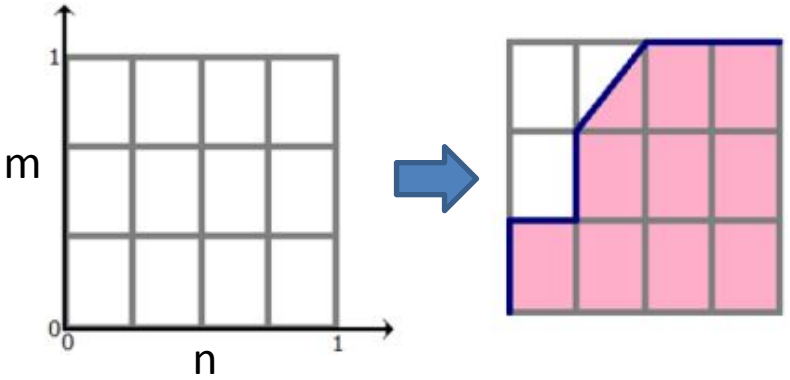
- take a unit square on the coordinate plane, divide it into **m** equal parts by horizontal lines and into **n** by vertical lines, where **m** - is the number of objects of class 1 among test set, **n** - is the number of 0 class in test set. As a result, the square is divided by a grid into ***m×n*** blocks.
- Starting from (0,0) and **going through sorted predictions in descending order;**
- If the actual label is 1, then we step up; if 0, then we take a step to the right.
- In the end we will get to the point (1, 1), because we take a total of **m** steps up and **n** steps to the right.
- If the predictions are the same for the set of objects than we step diagonally up-right.

Id	P(x)	Label
1	0.5	0
2	0.1	0
3	0.2	0
4	0.6	1
5	0.2	1
6	0.3	1
7	0.0	0

initial

Id	P(x)	Label
4	0.6	1
1	0.5	0
6	0.3	1
3	0.2	0
5	0.2	1
2	0.1	0
7	0.0	0

sorted



AUC = 9.5/12 = 0.79

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